

Portfolio Intelligence & Predictive Debt Recovery Analytics

Turning Portfolio Data into Smarter
Recovery Decisions

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Portfolio at a Glance

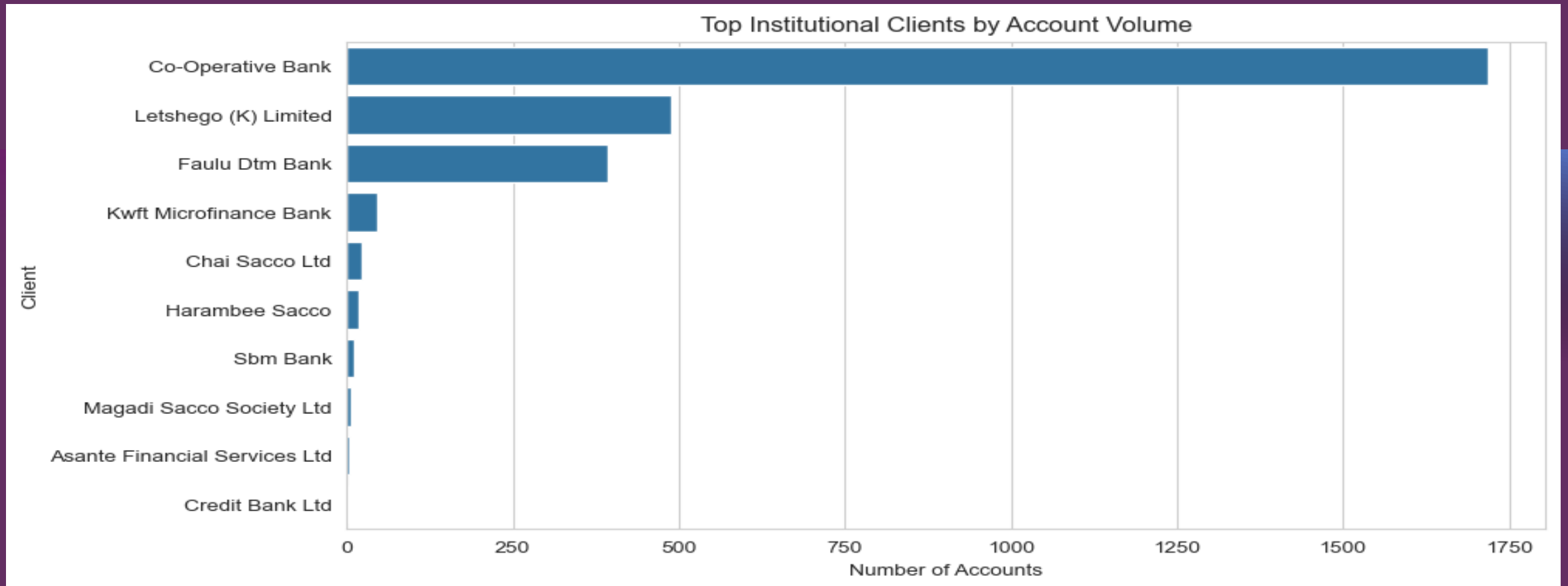
Key Question: *"What does the active portfolio look like today?"*

•6-KPI Executive Grid:

Metric Card	Value	Operational Note
Total Portfolio Exposure	KES 232.59M	Active outstanding balance across all clients
Active Accounts	2,709	Individual debtor files assigned to one agent
Average Account Balance	KES 85,821	Mean principal ticket size
Average Delinquency (DPD)	114 Days	Portfolio-wide Days Past Due average
High-Risk Exposure Volume	684 Accounts	Accounts in late-stage default (>90 DPD)
Active Client Institutions	4 Institutions	Financial partners (Banks, Microfinance, SACCOs)

Portfolio Composition & Exposure Distribution

Key Question: Where is our money concentrated, and which client accounts drive volume vs. value?



Key Strategic Insights: Client Portfolio Composition

Extreme Volume Concentration: Co-Operative Bank dominates operations with 60%+ of all active accounts (~1,720+).

Top 3 Core Drivers: Co-Op, Letshego, and Faulu DTM collectively represent ~95% of total portfolio volume.

Fragmented Long-Tail: 7 out of 10 clients (SACCOs & micro-lenders) hold minor volumes (<50 accounts each).

Operational Strategy

Dedicated Co-Op : Assign a specialized collector team to Co-Op to prevent workflow bottlenecks.

Long-Tail Automation: Route low-volume SACCO portfolios into automated digital outreach to save agent capacity.

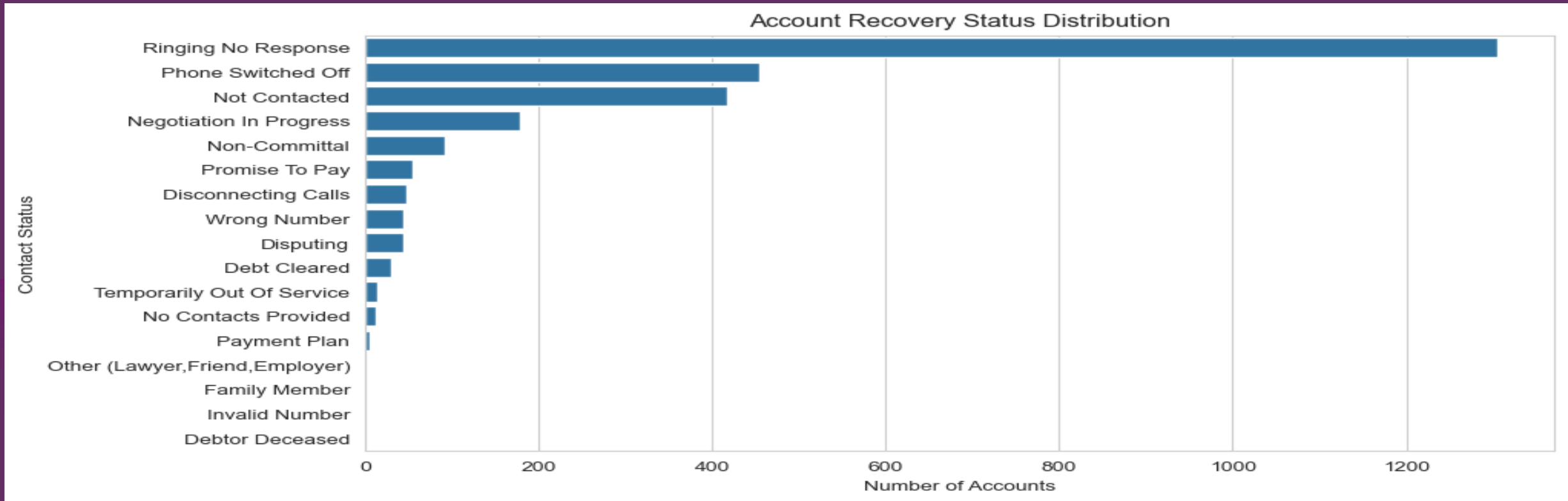
Exploratory Data Analysis: Key Behavioral Insights

Key Question: *What underlying behavioral patterns exist in our portfolio before we apply machine learning?*

💡 *Not all delinquent accounts behave similarly. Portfolio analytics identifies patterns that enable targeted recovery strategies before predictive models are applied.*

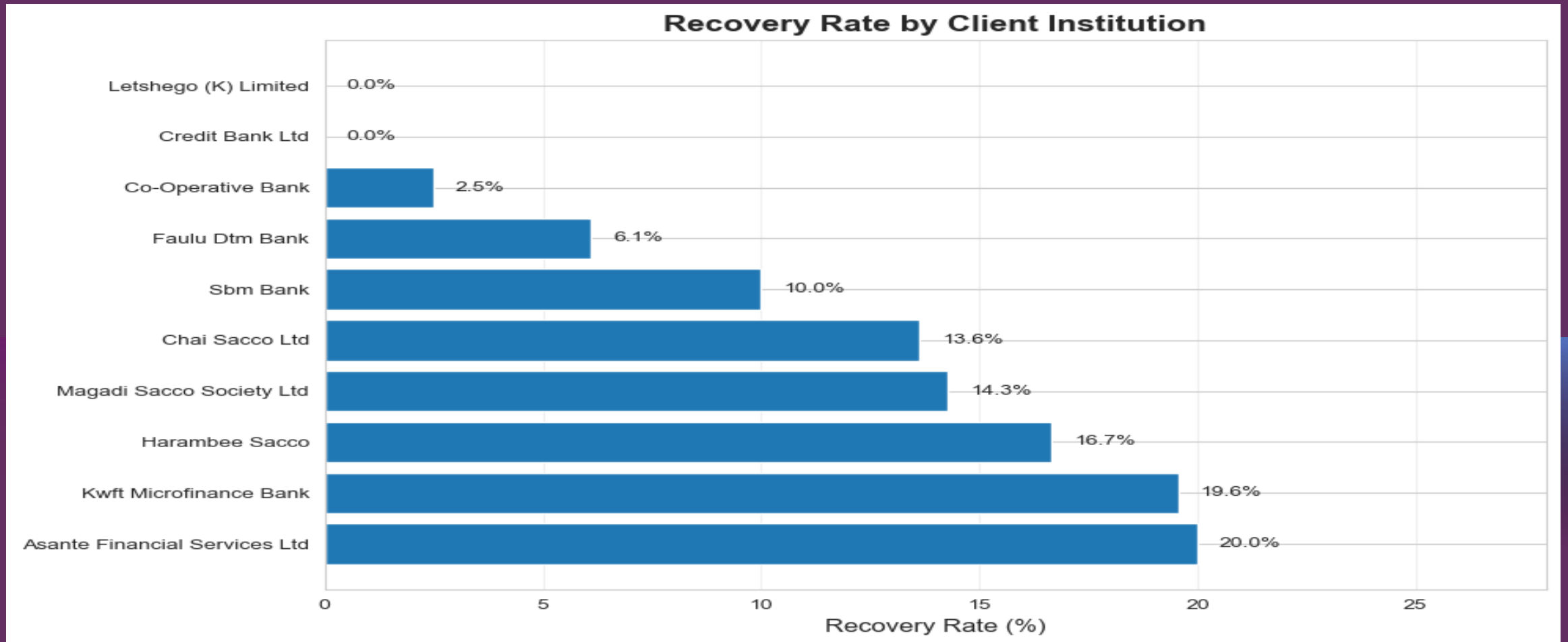


Contact status distribution



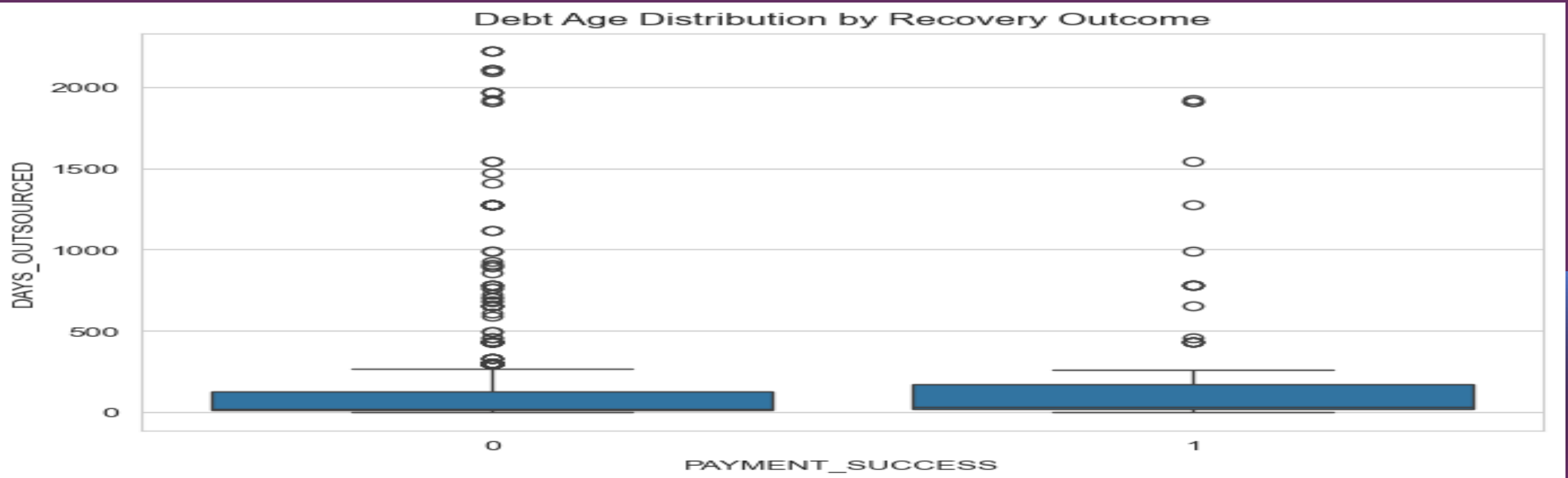
Contact Status Distribution: Reachable accounts convert at a significantly higher rate regardless of DPD.

Recovery rate by client institution

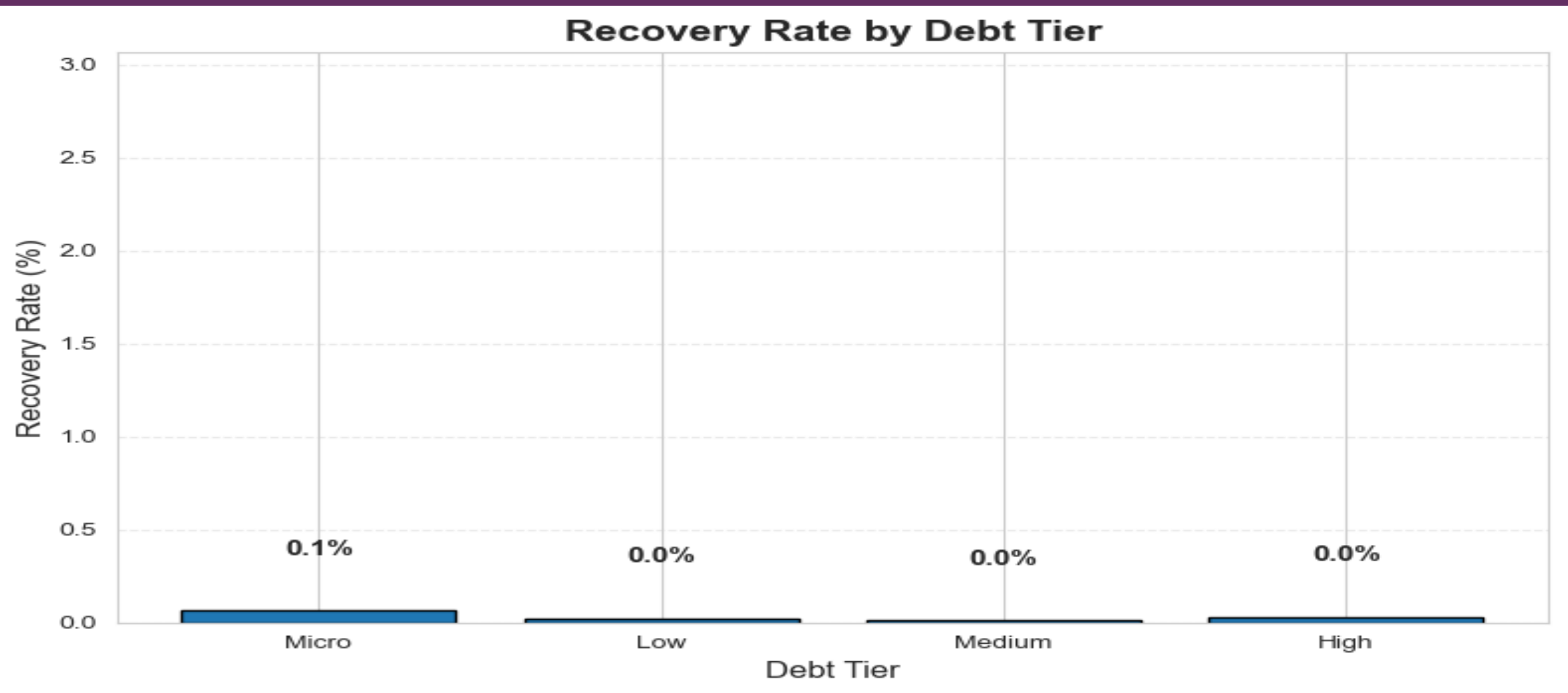


Recovery Rate by Client Institution: Distinct recovery benchmarks across different lending products.

Debt Age Distribution by Recovery Outcome



Debt Age (DPD) vs. Payment Success: Sharp drop-off in voluntary repayment past the 90-day threshold.



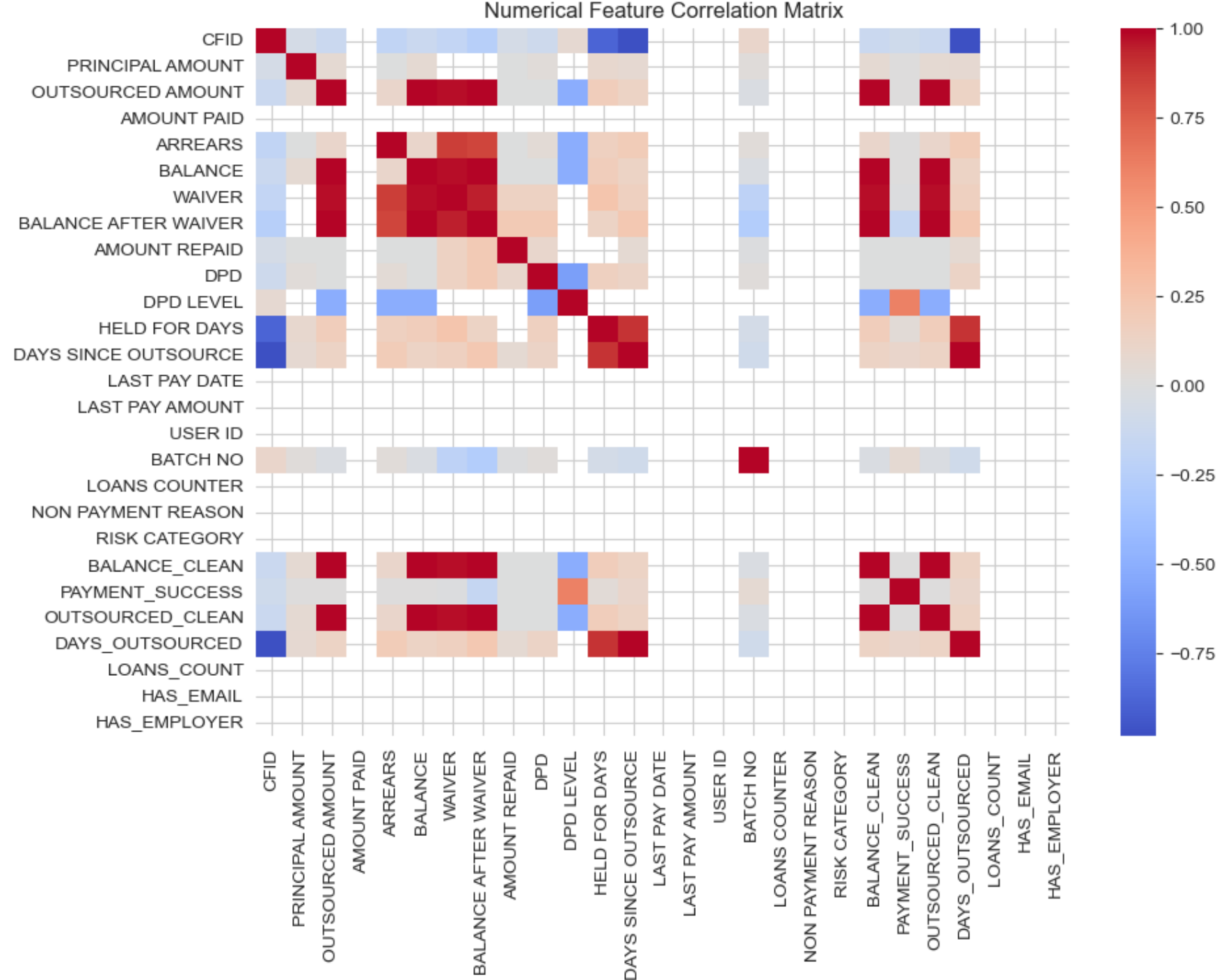
Recovery Probability by Debt Tier: High-balance tiers exhibit different repayment elasticities compared to low-tier balances.

Portfolio Risk Analysis & Heatmap

Risk Concentration: Largest monetary exposure lies within 90+ DPD accounts.

High-Value Urgency: Large-balance accounts require immediate, senior collector intervention.

Non-Linear Risk: Portfolio risk is concentrated in specific aging segments rather than evenly distributed.



Recovery Prioritization Matrix

Key Question: "How should operational teams triage accounts for maximum recovery yield?"

Category	Segment Criteria	Recommended Operational Action
 Immediate Recovery	High Balance + High Reachability	Senior Agent Direct Outreach & Immediate PTP Tracking
 Automated Follow-up	Low Balance + Early DPD (0–60)	Digital Outreach (Automated SMS & M-Pesa Payment Prompts)
 Field Investigation	High Balance + Low Reachability / High DPD	Skip Tracing, Workplace Tracing & Field Agent Dispatch
 Legal Escalation	Low Recovery Score + 180+ DPD	Pre-Legal Demand Notices & Formal Write-off Review

Collector Performance & Operational Metrics

Key Question: *What metrics can we measure today, and what will we track as operational data matures?*

Metrics Comparison Table

Category	 Current Capabilities (Proven)	 Future Metrics (Phase 2)
Workload & Allocation	Assigned Accounts: Total account volume per agent	Average Handle Time: Call duration & wrap-up speed
Commitments & SLAs	Tracked PTP Volume: Promised collection value	SLA Adherence: Timeliness of scheduled PTP follow-ups
Financial Recovery	Collection Yield: Absolute value recovered (KES)	Call-to-Collection Rate: Conversion efficiency per call
Contact Efficiency	Baseline Reachability: Contact success rate (%)	Agent Reach Scores: Granular contactability rating

Machine Learning Pipeline & Explainability Engine

Key Question: How does machine learning turn raw portfolio data into actionable daily priorities?

1. PORTFOLIO DATA INGESTION

Inputs: Accounts (KES 232.59M portfolio total)

Features: Days Past Due (DPD), Outstanding Principal, Historical Payment Behavior

Objective: Feed multi-source credit variables into structured pipeline



2. PREDICTIVE SCORING

(XGBoost)

Algorithms:

XGBoost Classifier & Random Forest

Class Balancing:

Optimized for highly imbalanced recovery outcomes

Accuracy: ROC-AUC ~0.82+ across delinquent portfolio evaluation folds



3. SHAP EXPLAINABILITY LAYER

Transparency:

Breaks down exact feature contributions per account

Primary Drivers:

Highlights DPD and Principal Owed as top risk factors

Governance:

Eliminates "black-box" decisions for executive oversight



4. DYNAMIC QUEUE PRIORITIZATION

Tiered Output:

Assigns High, Medium, or Low Recovery Probability

Agent Routing:

Automatically routes high-yield accounts to collectors first

ROI Impact:

Minimizes wasted field agent hours on unrecoverable debt

Executive Dashboard Architecture (Streamlit UI)

Key Question: What operational interface will management and agents interact with daily?

4-Panel Interface Overview:

1. **Executive Overview Panel:** High-level KPIs, exposure charts, and client filtering.
2. **Prediction Engine Page:** Machine learning scoring interface displaying account-level probability scores.
3. **SHAP Model Explanation View:** Visual breakdown of features driving individual account predictions.
4. **Collector Priority Queue:** Daily action lists sorted by recovery score and potential financial yield.

Streamlit Screenshots



Keysian Recovery Ops

Keysian Auctioneers & Debt Recovery Unit

KEYSIAN DEBT RECOVERY UNIT

Portfolio Analytics & Operations Intelligence Platform

Total Debt Portfo...
10 Acc...

Total Portfolio Expos...
KES 2,...

Collected (Last 30 D...
KES 3...
↑ 13.29% Rec. Rate

30-Day Recovery Rate
13.29%

Avg Account Balance
KES 2...

Portfolio Aging Analysis & Recovery Performance

Aging Bucket Summary Breakdown Exposure Share by Aging Category

Active Portfolio Dataset

Upload Excel or CSV Portfolio

Upload

200MB per file • XLSX, XLS, CSV

Navigation

- Executive Overview & Alerts
- Full Allocated Portfolio Directory
- Allocate New Portfolio Batch
- Collector Performance Portal
- AI Skip Tracing
- Smart Call Center & Auto Notes
- Automated Comms & M-Pesa Link
- Optimal Call Timing Predictions
- Printable Debt Cards & Export

Manage app

Active Portfolio Dataset

Monitor individual agent performance, call counts, and payment collection adherence.

Select Collector / Recovery Agent

Charity Nduati

Assigned Accounts
48

Calls Made Today
37

Total Collected This Month
KES 1,240,000

Active Accounts Assigned to Charity Nduati

	Account ID	Debtor Name	Client Institution	Phone Number	Email Address	Outsourced Amount (KES)	Amount Co
0	KEY-2026-001	Wanjiru Kinuthia	Faulu Microfinance	+254712345678	wanjiru@example.co.ke	185000	
6	KEY-2026-007	David Omwamba	Equity Bank	+254778901234	david@example.co.ke	140000	

Navigation

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Manage app

Summary of Streamlit Dashboard

Business Impact: Before vs. After Transformation

Operational Capability	Traditional Debt Recovery	Enterprise Portfolio Intelligence
Portfolio Oversight	Static Excel spreadsheets & manual reviews	Real-time interactive dashboards & dynamic filtering
Account Prioritization	Intuition-based / manual agent selection	Machine learning risk & probability scoring
Resource Allocation	Equal allocation across collectors	Priority-based assignment by balance & risk tier
Operational Strategy	Reactive follow-ups after default occurs	Predictive reachability & targeted intervention
Transparency & Audit	Fragmented collector notes	Standardized analytics & transparent SHAP explanations

Strategic Recommendations & Executive Conclusion

Translating Portfolio Analytics into Actionable Operations

 1. WORKLOAD ALLOCATION	 2. DIGITAL AUTOMATION	 3. HIGH-VALUE FOCUS
Co-Op Bank	SACCO Long-Tail	Priority Scoring
• Assign dedicated collector team for 60%+ volume	• Consolidate low-volume clients (<50 accounts)	• Assign top-tier exposure (>90 DPD) to senior agents
• Prevents operational bottlenecks	• Saves agent capacity for high-yield loans	• Maximizes cash collection yield

Thank you

Questions?

